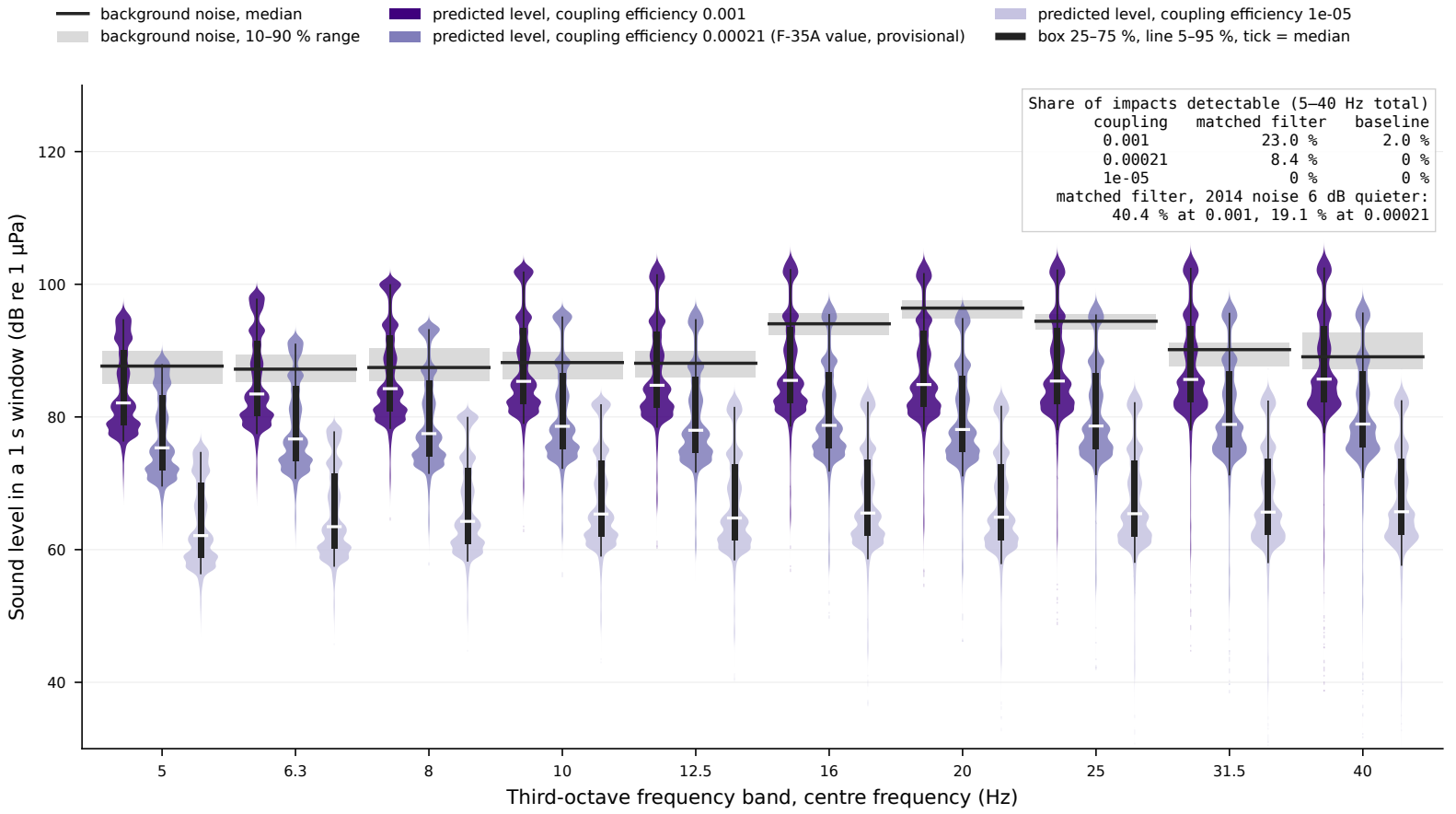
Cape Leeuwin (H01W): background noise against the predicted sound level of all simulated impacts, by frequency band



Plain note: The grey line is the median background noise at the station in each frequency band. The grey box holds 80 % of the noise levels. Each coloured shape shows all the simulated impacts: its width shows how many impacts give that sound level at the station. There is one shape for each of three assumed shares of impact energy that become sound. Where a shape is above the grey line, those impacts are louder than the noise in that band. The box at the top right gives the share of impacts that two detectors would find in the 5 to 40 Hz total. Below 5 Hz there is no prediction. These are estimates. They are not calibrated.

Technical note: Source posterior: end-of-flight run J (74 seeds, 313.6 M rows, four strata pooled by p_core/K) + northern supplement (impacts-ruleB.npz); weighted subsample 2 M draws per 00:19 option at dive probability p = 0.5 (run_j_levels.py, prereg 4277320). Panel: 00:19 option D0′ (held out), p = 0.5; Kish ESS of the subsample 1.90 M (Kish ESS of the full run J mixture for D0′ at p = 0.5: 1.34 M, run-j-windows/ess.json). E = contact kinetic energy (upper anchor), tau = tau90 clamped to 0.05–10 s, 10 m source, one-pole spectrum flat to 1/tau and f^-2 above; level = 1 s-equivalent third-octave band SPL (T_eff 2.68 s), dB re 1 uPa; eta only shifts the distribution by 10 log10 eta. Violins: weighted histograms of the 2 M draws (0.25 dB bins, 0.5 dB Gaussian smoothing), width normalised per violin; mass below 30 dB is not drawn. TL: whole-arc KRAKEN grid (offset-0 line, latitude-only, clamped 41–20 S); IMOS transmission loss: corrected RAM-along-the-arc table imos_tl_ram_arc.csv (imos_ram_grid.py, run_j_levels amendment 2; replaces imos_tl_ramcorr_arc.csv); IMS (H01W/H08S/H08N) loss unchanged. RAM delta at H01W/H08S, none at H08N; 3250 uncorrected. Noise: Blackman et al. (2004) App. B, May-June 2003, digitised (PSD x band width). Detectable = SNR of the 5–40 Hz total >= 7 dB (envelope matched filter, P_D 0.5, alpha 0.01) or >= 15 dB (STA/LTA baseline), model-matched upper bounds from the weak-signal trial on 3376 background; SNR against the module's 5–40 Hz noise medians (imos2b; IMS = 3376 proxy + Blackman offset, +4.8 dB triad gain). The 2014 line uses Kadri Fig. 9 panel (b) noise, 5.96 dB below Blackman 2003 (digitised, +/-2 dB). PREDICTIVE, NOT CALIBRATED. C1: propagation slope misfit +10.4 / +7.4 dB per octave at H01W / H08S (after the EW0112 source result a source-term caveat; air9 verdict NOT EXPLAINED, gap G23). C2: coupling efficiency eta not calibrated; swept over four decades; F-35A upper anchor eta = 2.1e-4 is provisional (from Brown 2026's quote of Metz 2023; re-derivation from Metz pending). No detection is claimed.